

Class test.. science (physics)

THE MATHEMATICS TUTORIAL

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1. In the following cases, a ray is incident on a concave mirror. In which case is the angle of incidence equal to zero?(1)

- (A) A ray parallel to the principal axis.
- (B) A ray passing through the centre of curvature and incident obliquely.
- (C) A ray passing through the principal focus and incident obliquely.
- (D) A ray incident obliquely to the principal axis, at the pole of the mirror

2. When an object was kept at position X in front of a concave mirror, an enlarged and virtual image was formed. Which among the following identifies 'X' correctly?(1)

- (A) Anywhere between the centre of curvature and principal focus.
- (B) Anywhere between the pole and principal focus.
- (C) Exactly at the centre of curvature.
- (D) Exactly at the principal focus

3 Consider the following properties of virtual images:(1)

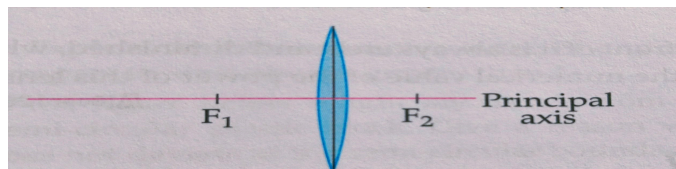
- (i) cannot be obtained on the screen.
 - (ii) are formed by both concave and convex lens.
 - (iii) are always erect.
 - (iv) are always inverted.
- The correct properties are:
(A) (i) and (iv) (B) (i) and (ii) (C) (i), (ii) and (iii) (D) (i), (ii) and (iv)

4. The refractive index of medium A is 1.5 and that of medium B is 1.33. If the speed of light in air is 3×10^8 m/s, what is the speed of light in medium A and B, respectively?(1)

- (A) 2×10^8 m/s and 1.33×10^8 m/s
- (B) 1.33×10^8 m/s and 2×10^8 m/s
- (C) 2.25×10^8 m/s and 2×10^8 m/s
- (D) 2×10^8 m/s and 2.25×10^8 m/s

5. At what distance from a convex lens should an object be placed to get an image of the same size as that of the object on a screen?(1)

- (A) Beyond twice the focal length of the lens.
- (B) At the principal focus of the lens.
- (C) At twice the focal length of the lens.
- (D) Between the optical centre of the lens and its principal focus.



6. The above image shows a thin lens of focal length 5m.(3)

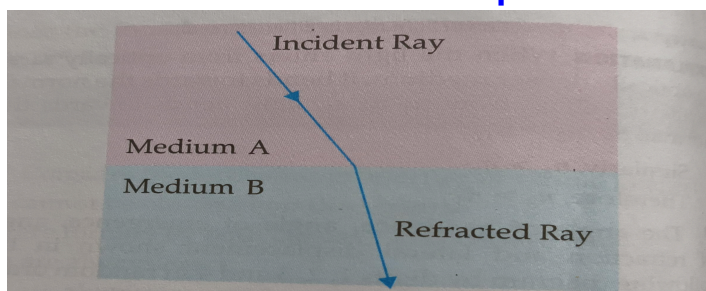
- (i) What is the kind of lens shown in the above figure?
- (ii) If a real inverted image is to be formed by this lens at a distance of 7m from the optical centre, then show with calculation where should the object be placed?
- (iii) Draw a neatly labelled diagram of the image formation mentioned in (ii).

7. Refractive index of water with respect to air is 1.33 and that of diamond is 2.42. (3)

- (i) In which medium does the light move faster: water or diamond?
- (ii) What is the refractive index of diamond with respect to water?

8. A light ray enters from medium A to medium B as shown in the figure (3)

- (a) Which one of the two media is denser w.r.t. other medium? Justify your answer.
- (b) If the speed of light in medium A is V_A and B is V_B , what is the refractive index of B with respect to A?



9. (a) A security mirror used in a big showroom has radius of curvature 5 m. If a customer is standing at a distance of 20 m from the cash counter, find the position, nature and size of the image formed in the security mirror. (3)

(b) Neha visited a dentist at the dentist's clinic. She observed that the dentist was holding an instrument fitted with a mirror. State the nature of this mirror and reason for its use in the instrument used by dentist.

Q.10. A student focussed the image of a candle flame on a white screen using a convex lens. He noted down the position of candle, screen and the lens as under - (3)

Position of candle = 12 cm

Position of convex lens = 50 cm

Position of screen = 88 cm

(a) What is the focal length of convex lens ?

(b) Where will the image be formed if the further shifts the candle towards the lens at a position of 31.0 cm ?

(c) What will be the nature of image formed if he further shifts the candle towards the lens ?

(d) Draw a ray diagram to show the formation of image in (c) above.

OR

11. Draw well level diagrams of concave mirror and convex lens when

(a) images is formed virtual and erect

(b) Object place on C (centre of curvature) and $2F_1$